A Multi-Mode Simulation Framework for X-Band Pulse Compression Radar

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Motivation: Data Scarcity for Radar-Based Perception

Problem:

- Autonomous USVs require robust radar perception
- Deep learning models need large labeled datasets
- Field data collection is expensive and time-consuming
- Small recreational boats produce intermittent, low-RCS returns
- Inland waterway clutter is heterogeneous and poorly characterized

Why X-Band Marine Radar?

- All-weather, day-night operation
- Relevant collision avoidance ranges
- COTS units are compact, affordable, field-proven
- Complementary to camera/lidar

Solution:

- Generate synthetic training data
- Multiple abstraction levels for different workflow stages

Related Work and Background

Synthetic Data for Perception

- Domain randomization effective for camera-based object detection
- SAR simulation well-developed, but navigation-band marine radar lags
- Sim-to-real transfer remains challenging for radar

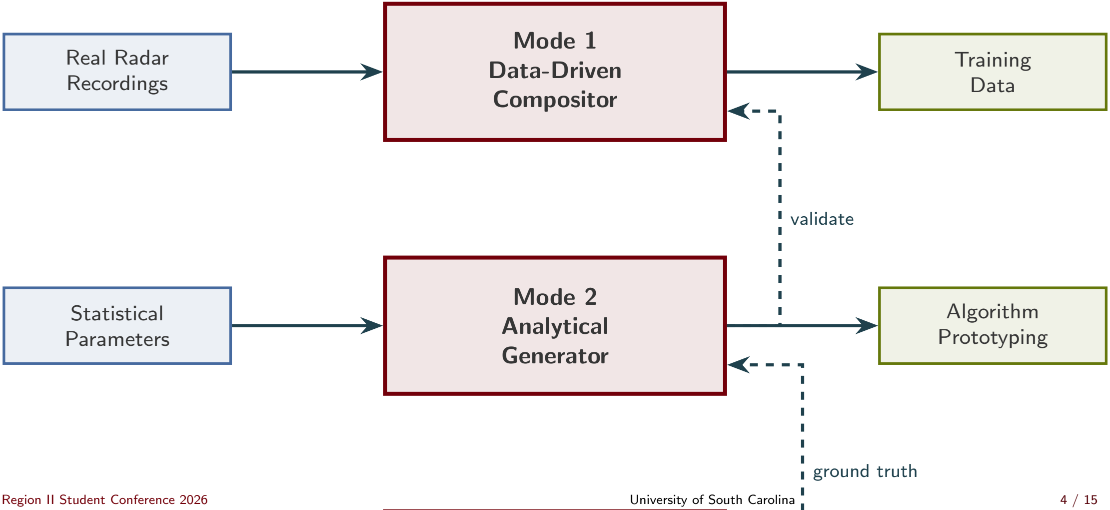
Sea Clutter Modeling

- Compound K-distribution: standard model for X-band sea clutter
- Captures heavy-tailed amplitude statistics from Bragg scattering
- Ocean models available, but inland waterways poorly characterized

Marine Radar Object Detection

- Traditional: CFAR processors
- Modern: Deep learning (YOLO, Faster R-CNN)

Multi-Mode Framework Architecture



Key Advantages:

- Inherits authentic sensor noise, clutter texture, and artifacts
- Minimal sim-to-real gap
- Complete control over target placement and labeling

Environmental Augmentation:

- Radar interference patterns extracted from Charleston Harbor
- Rain clutter textures from weather events
- Exposes detector to degraded conditions without additional field collections

Approach: Blend synthetic target signatures into real radar recordings

Generation; [step, right=0.3cm of s2] (s3) 3. Amplitude
Matching; [step, right=0.3cm of s3] (s4) 4. Insertion
and Blending; [step, right=0.3cm of s4] (s5) 5. Label
Generation;
[arrow] (s1) – (s2); [arrow] (s2) – (s3); [arrow] (s3) – (s4); [arrow] (s4) – (s5);

Mode 2: Analytical Generator

Approach: Synthesize complete radar frames from parametric statistical models

Clutter Model:

Compound K-distribution

$$f(x) = \frac{4}{\Gamma(\nu)} \left(\frac{\nu}{P_c}\right)^{\frac{\nu+1}{2}} x^{\nu}\, K_{\nu-1}\left(2\sqrt{\frac{\nu\, x^2}{P_c}}\right)$$

- ν: shape parameter (tail behavior)
- P_c: mean clutter power
- Small ν = heavier tails, spikier clutter
- Inland lakes: large ν (approaching Rayleigh)

Target Model:

Fluctuating point scatterers, Swerling I distribution

Radar equation:

$$P_r = \frac{P_t\, G^2\, \lambda^2\, \sigma}{(4\pi)^3\, R^4\, L}$$

Frame Synthesis:

1. Draw clutter realization from K-distribution
2. Add target returns with Swerling RCS
3. Apply log-detection
4. Add thermal noise and interference

Mode 3: EM Ray Tracing

Approach: Shooting-and-bouncing-rays (SBR) for pulse-level EM simulation

Architecture:

1. Launch rays from radar antenna (beam pattern)
2. Propagate through scene, intersect water surface and targets
3. Generate reflected/scattered rays (material-dependent)
4. Coherently sum rays returning to aperture
5. Match-filter (pulse compression) and log-detect

Water Surface:

- Time-varying mesh from wind-driven wave spectrum

Purpose:

- Physics-grounded reference data
- Validate assumptions in Modes 1 and 2
- Study multipath, scattering, and clutter mechanisms

Current Status:

- Functional for canonical scenarios (single targets, flat/perturbed water)
- Preliminary validation against two-ray (Lloyd's mirror) model
- Extension to complex scenes (shorelines, docks, vegetation) ongoing

Sensor Platform: Furuno DRS4D NXT

Parameter	Value
Transmitter	Solid-state, pulse compression
Frequency	9.38–9.44 GHz (X-band)
Output power	100 W
Modulation	LFM chirp / unmodulated
Chirp width	5.0–18 μs (range-dependent)
Range resolution	$\sim$20 m
Horiz. beamwidth	3.9° (RezBoost disabled)
Vert. beamwidth	22–25°
Rotation rate	24 / 36 / 48 RPM
PRF	700–1100 Hz
Azimuth quant.	8,192 spokes/rev
Radome diameter	610 mm (24 in)

Key Features:

- Doppler-capable
- RezBoost disabled to preserve weak returns
- UDP multicast spoke-format intensity data
- Log-detected, non-coherent (no I/Q)

Target RCS Estimates:

- Fiberglass boats: 0.5–20 m^2
- Aluminum boats/buoys: 1–50 m^2

Data Collection Campaign

Locations:

- Lake Murray, SC
- Lake Greenwood, SC
- Charleston Harbor, SC

Duration:

- 11 months (Jan–Nov 2025)
- $\sim$30 sessions
- $>$XXX,000 frames

Conditions:

- Fair weather (majority)
- 1–2 rain events
- Radar interference (Charleston)

Geometry:

- Shore-based, fixed installations
- Antenna height: 2–4 m above water
- Range scales: 0.25–0.75 NM

Extreme Low Grazing Angle:

At $h = 3$ m, $R = 926$ m (0.5 NM):

$$\psi = \arctan\left(\frac{h}{R}\right) = \arctan\left(\frac{3}{926}\right) \approx 0.19^\circ$$

- Multipath interference lobes
- Targets fade in/out
- Drives intermittent-return tracking challenges

Results: Domain Transfer for Object Detection

Experiment: Train YOLO detector exclusively on Mode 1 synthetic data, evaluate on real radar

Training Configuration:

- Training data: TBD composite B-scan frames (Mode 1)
- Annotations: Automatically generated bounding boxes
- Model: YOLOvX (TBD)
- No real data used during training

Evaluation:

- Test set: Real radar frames with annotated targets
- Targets: Recreational boats and navigational buoys

Table: Detection performance on real data

Training	Test	mAP@0.5
Mode 1 (clean)	Real (clean)	TBD

Key Finding:

- Synthetic training data enables effective domain transfer
- Mode 1 compositor minimizes sim-to-real gap by inheriting authentic sensor characteristics

Results: Robustness Under Degraded Conditions

Experiment: Augment Mode 1 training data with real interference and rain-clutter textures

Table: Augmentation experiment results (mAP@0.5)

Training Set	**Test Set**	**mAP@0.5**
Clean Mode 1	Clean real	TBD
Clean Mode 1	Rain real	TBD
Clean Mode 1	Interference real	TBD
Augmented Mode 1	Rain real	TBD
Augmented Mode 1	Interference real	TBD

Augmentation Strategy:

- Extract real interference patterns from Charleston Harbor recordings
- Extract rain-clutter textures from weather events
- Overlay onto clean Mode 1 composite frames

Discussion and Key Insights

Inland Waterway Characteristics

- Differs substantially from open-ocean environments
- Near-glassy calm with light wind chop
- Heterogeneous shoreline returns and man-made structures
- K-distribution ν expected large (approaching Rayleigh)
- Inland-waterway radar perception underrepresented in literature

Low Grazing Angle Effects

- Grazing angle $< 0.5^\circ$ throughout scene
- Strong multipath interference (two-ray pattern)
- Targets fade through constructive/destructive lobes
- Drives intermittent-return behavior and track fragmentation

Mode 1 Compositor Strengths

Limitations and Future Work

Current Limitations

- Mode 2 analytical generator validation against real clutter statistics ongoing
- Mode 3 ray tracer does not yet model complex shoreline geometry or vegetation
- Study limited to fair-weather inland conditions
- Training/test split uses data from same collection sites
- Cross-site generalization not yet evaluated

Planned Extensions

- Complete Mode 2 validation with full clutter characterization
- Extend Mode 3 to heterogeneous inland scenes
- Evaluate cross-site generalization (train on Lake Murray, test on Charleston)
- Expand to higher sea states and open-water conditions
- Integrate tracking algorithms for intermittent-return management

Conclusion

Summary:

- Presented multi-mode simulation framework for X-band pulse compression radar
- Three modes organized by workflow utility:
  - Mode 1: Data-driven compositor for production training data
  - Mode 2: Analytical generator for rapid algorithm prototyping
  - Mode 3: EM ray tracing for physics validation
- Demonstrated YOLO detector trained exclusively on Mode 1 synthetic data achieves TBD mAP@0.5 on real radar frames
- Augmenting synthetic data with real degradation patterns improves robustness

Contributions:

- Multi-mode framework tailored to COTS marine radar in inland/littoral environments
- Effective domain transfer for deep learning-based radar object detection
- Augmentation strategy using real interference and rain-clutter textures
- Detailed characterization of Furuno DRS4D NXT as research sensor

Impact: Controlled, reproducible environment for developing radar-based USV perception without extensive labeled field collections

Questions?

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